

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 01 · LECTURE NOTES

What a Camera Measures

The measurement, and everything that is not in it.

MODULE OUTCOMES

- Describe image formation from scene radiance to stored value.
- Explain the exposure triangle in terms of what each control does to the measurement.
- Explain what a raw file contains and what it does not.
- Explain noise sources and why they behave as they do.
- Explain why linearity matters and where it is lost.

1 Image formation

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scene radiance  $L(x,y,\lambda)$     [W/m2/sr]
  | lens: aperture, focal length, vignetting, aberration
image irradiance  $E(x,y,\lambda)$  [W/m2]
  | shutter: integrate over exposure time
photon count per well    [electrons]
  | colour filter array; quantum efficiency; full well
analogue voltage
  | analogue gain (ISO)
ADC  ->  RAW INTEGER    [12-14 bits, LINEAR]
  | demosaic, white balance, colour matrix, tone curve
display image           [8 bits, NON-LINEAR]

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The raw integer is proportional to photon count. That proportionality is the measurement, and it is the only thing in the pipeline that has a defensible claim to being what the camera saw. **Everything below the ADC line is interpretation** — a series of choices about how to turn a measurement into something that looks right on a screen, and this course is largely about those choices.

Control	What it changes	Side effect
Aperture	How much light is gathered per unit time. A wider aperture collects more.	Depth of field narrows as the aperture widens. And at very small apertures, diffraction softens the image — so stopping down past a point loses sharpness rather than gaining it.
Shutter speed	How long the sensor integrates. Twice the time, twice the photons.	Motion blur , from subject movement or camera shake.

Control	What it changes	Side effect
ISO	Analogue gain applied after the measurement. It does not change how many photons were collected.	Amplifies signal and noise together , so the signal-to-noise ratio is unchanged or slightly worsened. High ISO is noisy because it is used in situations where few photons were available, not because the gain creates noise.

ONLY TWO CONTROLS CHANGE THE PHOTON COUNT

Aperture and shutter time determine how many photons reach the sensor. ISO does not. This is the most commonly misunderstood fact in photography, and it has direct consequences for every computational technique in this course.

Photon count determines the noise floor (§3), so image quality in the sense that matters — signal-to-noise — is fixed by aperture and shutter alone.

ISO is a decision about where to place the measurement within the converter's range. It is genuinely useful — raising gain before the ADC can keep a dim signal above the quantisation floor and the read noise — and it is not a way to obtain more signal.

This is why 'expose to the right' is sound advice: collect as many photons as the highlights will tolerate without clipping, because photons you did not collect cannot be recovered afterwards by any amount of processing. A dark raw file brightened later is noisier than a bright raw file darkened later, and the difference is not subtle.

2 The raw file

	Raw	JPEG
Values	Linear in scene radiance, 12–14 bits per pixel.	Non-linear — gamma or sRGB encoded — 8 bits.
Colour	One channel per pixel , in the colour filter array's pattern (Module 05).	Three channels per pixel, already interpolated.
White balance	Not applied — recorded in metadata as multipliers you may use or ignore.	Baked in, irreversibly. Correcting it later means scaling already-clipped, already-encoded values.
Also contains	Black level, saturation point, the sensor's colour matrix, and the CFA pattern — everything needed to interpret the numbers.	An embedded profile, and the camera manufacturer's opinion about sharpening and noise reduction.
In short	The measurement.	An interpretation.

LINEARITY IS THE PROPERTY EVERYTHING DEPENDS ON

A raw value of 2000 corresponds to exactly twice the light of a value of 1000. The relationship is proportional, by construction, because the sensor counts photons and the ADC is linear.

So raw values can be added, averaged, and scaled meaningfully. Merging exposures into a radiance map (Module 03), averaging a burst to reduce noise (Module 10), and every physically-motivated operation in this course require exactly this property.

A gamma-encoded JPEG value does not have it. Averaging two gamma-encoded pixels does not give the encoding of the average radiance — the result is systematically wrong, and for high-contrast pairs it is badly wrong. Half of a bright highlight and half of a shadow is not what you get.

So: do all processing in linear space, and encode for display at the very end. This is the identical rule that CSCE 647's Module 13 arrived at from the rendering direction — the two courses meet here, and for the same reason.

3 Noise

$$\text{shot noise: } \sigma = \sqrt{N}$$

$$\text{SNR} = N / \sqrt{N} = \sqrt{N}$$

Source	Behaviour	Dominant when
Shot noise	Photon arrival is a Poisson process, so the standard deviation of a count N is $\sqrt{N}$. A property of light itself.	Essentially always, in normal photography.
Read noise	Added by the readout electronics and the ADC. Roughly constant regardless of signal.	Very dim scenes, where the photon count is small enough that a constant additive term matters.
Dark current	Thermally generated electrons, proportional to exposure time and strongly dependent on temperature.	Long exposures, and warm sensors. Astrophotography subtracts a 'dark frame' for exactly this.
Fixed pattern noise	Per-pixel variation in gain and offset. Static, so it can be calibrated out.	Visible mainly after heavy amplification.

SHOT NOISE IS WHY LOW LIGHT IS HARD, AND IT IS NOT THE SENSOR'S FAULT

Photons arrive as a Poisson process. If the expected count in a well is N , the standard deviation of the actual count is $\sqrt{N}$. This is a statement about light, not about any particular sensor.

In bright light N is large and the relative noise is small. Ten thousand photons gives a signal-to-noise ratio of 100, and the noise is invisible.

In dim light N is small and the relative noise is large. A hundred photons gives an SNR of 10, and the noise is a substantial fraction of the image. Ten photons gives an SNR of about 3, at which point the measurement barely constrains anything.

No improvement in sensor technology removes this. A perfect sensor — one that detected every photon with no read noise whatsoever — would still produce this noise, because it is in the signal being measured. **The only remedy is collecting more photons:** a wider aperture, a longer exposure, a larger sensor, or **more frames** — which is the entire premise of Module 10 and of modern phone photography.

- **Averaging n frames improves SNR by $\sqrt{n}$.** The signal adds linearly and the independent noise adds in quadrature, so the ratio improves as the square root. Four frames is equivalent to one frame with twice the exposure — without the motion blur. Module 10.
- **Denoising is estimation under a known noise model** rather than general-purpose smoothing. Knowing the noise is Poisson-Gaussian with measurable parameters is what makes principled denoising possible at all.
- **Noise is signal-dependent.** It is larger in absolute terms in the highlights (more photons, so more $\sqrt{N}$) and larger in relative terms in the shadows (fewer photons, so worse SNR) — which is why shadows look noisy even though the highlights carry more noise.
- **Which means denoising must be applied in linear space,** before any tone curve. A tone curve lifts the shadows and compresses the highlights, destroying the simple relationship between signal level and noise

level that a denoiser depends on. **Denoising after gamma encoding is a common mistake** and produces visibly worse results for a reason that is easy to state.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Marc Levoy — Lectures on Digital Photography, lectures 1–6 (free)	Image formation, exposure, and sensors, from someone who built camera pipelines. The clearest treatment of §1 available free.
CMU 15-463 — Image formation and the camera pipeline (free)	The lecture notes and the raw-processing assignment, which is essentially Project 1's first requirement.
Emil Martinec — Noise, Dynamic Range and Bit Depth in Digital SLRs (free)	The clearest explanation of §3 for photographers, with measurements. The site also publishes measured sensor data.
Hasinoff — Photon, Poisson Noise (free chapter)	The noise model of §3 stated precisely, by one of the HDR+ authors.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Shoot the same scene at several ISO values with compensating shutter speeds. Confirm that the noise depends on the photon count rather than on the ISO setting.
- 2 Open a raw file with rawpy and examine the array directly: the bit depth, the black level, the saturation point, and the CFA pattern.
- 3 Plot a histogram of raw values for a scene and identify the black level and the clipping point.
- 4 Photograph a uniform grey surface at several exposure levels and plot measured variance against mean. **The slope gives the gain and the intercept gives the read noise** — this is how noise models are calibrated.
- 5 Confirm the $\sqrt{N}$ relationship from that plot and report the deviation.
- 6 Average 4, 16, and 64 frames of a static scene and measure the noise reduction. Compare against the predicted $\sqrt{n}$.
- 7 Demonstrate why averaging must happen in linear space: average two very different exposures in linear and in gamma space and compare against ground truth.
- 8 Take the same scene as raw and as JPEG, and attempt to correct a deliberate white balance error in both. Report what is recoverable.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Trace the pipeline from scene radiance to a stored raw value.
- 2 Where is the boundary between measurement and interpretation?
- 3 What does each of aperture, shutter, and ISO change, and which affect photon count?
- 4 Why does 'expose to the right' work?
- 5 Give four differences between raw and JPEG content.
- 6 Why does linearity matter, and what goes wrong when averaging encoded values?
- 7 Why is shot noise $\sqrt{N}$, and why can no sensor improvement remove it?
- 8 How does SNR scale with frame count, and why?
- 9 Why must denoising be applied in linear space?